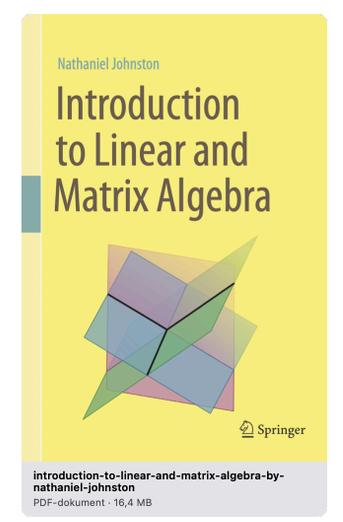

Nathaniel Johnston

Introduction to Linear and Matrix Algebra

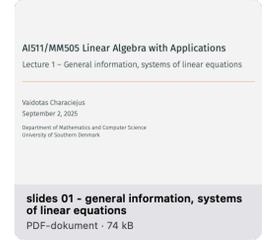
Springer

introduction-to-linear-and-matrix-algebra-by-nathaniel-johnston
PDF-dokument · 16,4 MB

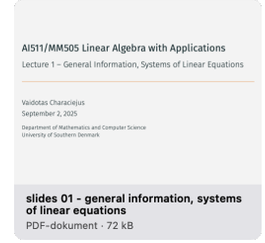
topics for this course

- Systems of linear equations.
- Matrix operations, inverses, determinants.
- Vector spaces, basis, coordinates, linear independence.
- Linear transformations, eigenvalue problems, diagonalisation.
- Inner product and orthogonality.

outdated


AI511/MM505 Linear Algebra with Applications
Lecture 1 – General information, systems of linear equations
Vaidotas Characiejus
September 2, 2025
Department of Mathematics and Computer Science
University of Southern Denmark

slides 01 - general information, systems of linear equations
PDF-dokument · 74 kB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Lecture 01

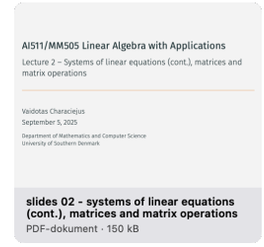
5 ECTS theory
– take home exam

2.5 ECTS Application
– explained by Henry

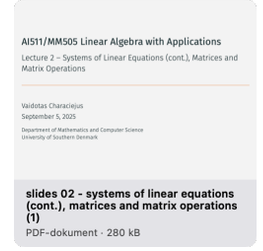
Systems of linear equations

use calculus to find linear equation (optimization)

outdated


AI511/MM505 Linear Algebra with Applications
Lecture 2 – Systems of linear equations (cont.), matrices and matrix operations

slides 02 - systems of linear equations (cont.), matrices and matrix operations
PDF-dokument · 100 kB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Lecture 2 – Systems of Linear Equations (cont.), Matrices and Matrix Operations

slides 02 - systems of linear equations (cont.), matrices and matrix operations (1)
PDF-dokument · 280 kB

Lecture 02

general system of linear equation...

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1; \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2; \\ \quad\vdots \qquad \vdots \qquad \vdots \qquad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m. \end{cases}$$

augmented matrix ↺

a_{ij} & b_i are entries

this can be rewritten to

$$\left[\begin{array}{cccc|c} x_{11} & x_{12} & \cdots & x_{1n} & b_1 \\ x_{21} & x_{22} & \cdots & x_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ x_{n1} & x_{n2} & & x_{nn} & b_n \end{array}\right]$$

now use elementary row operations (Gauss reduction)

Denotion

$\mathbb{R}^{m\times n}$

$(a_{ij})_{1\le i\le m,\ 1\le j\le n}$

Definition of an augmented matrix

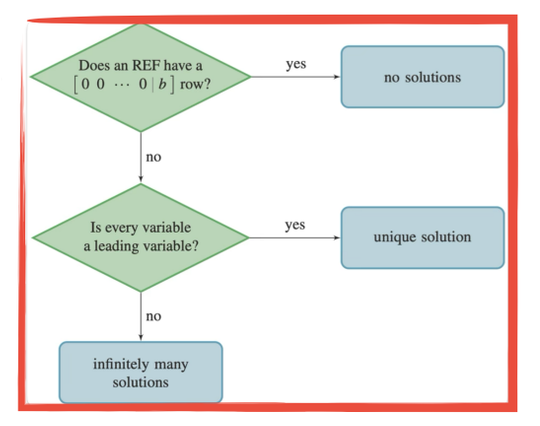
Definition
A (augmented) matrix is said to be in row echelon form (REF) if:
(a) all rows consisting entirely of zeros are below the non-zero rows;
(b) in each non-zero row, the first non-zero entry (called the *leading entry*) is to the left of any leading entries below it.

using Gauss-Jordan ↘ Definition of reduced REF

Definition
If a matrix that is in row echelon form also satisfies the following additional property, then it is in *reduced REF*:
(c) each leading entry equals 1 and is the only non-zero entry in its column.

if REF
use back substitution
elif reduced REF
read the answer

if the number of leading entries is less than the number of variables then there are infinite amount of solutions


Does an REF have a [0 0 ··· 0 | b] row?
yes
no solutions
no
Is every variable a leading variable?
yes
unique solution
no
infinitely many solutions

Matrix addition & scalar multiplication

Definition
Suppose that $A, B \in \mathbb{R}^{m\times n}$ and $c \in \mathbb{R}$. The *sum* $A + B$ and *scalar multiplication* cA are the $m \times n$ matrices whose (i, j)- entries, for each $1 \le i \le m$ and $1 \le j \le n$, are

$$(A + B)_{ij} = a_{ij} + b_{ij} \quad\text{and}\quad (cA)_{ij} = ca_{ij},$$

respectively.

↓ common rules for matrix addition & scalar multiplication

Theorem (Theorem 1.3.1 of Johnston (2021))
Suppose that $A, B, C \in \mathbb{R}^{m\times n}$ and $c, d \in \mathbb{R}$. Then
(a) $A + B = B + A$ (commutativity);
(b) $(A + B) + C = A + (B + C)$ (associativity);
(c) $c(A + B) = cA + cB$ (distributivity);
(d) $(c + d)A = cA + dA$ (distributivity);
(e) $c(dA) = (cd)A$.

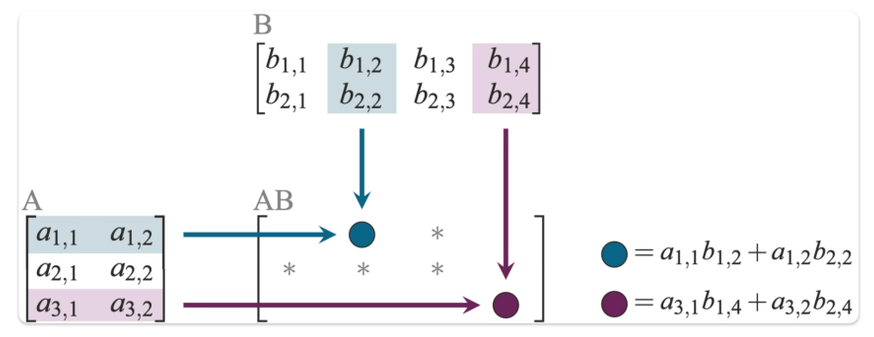
Matrix multiplication

Definition
If $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times p}$, then their *product* $AB \in \mathbb{R}^{m\times p}$ whose (i, j)-entry, for each $1 \le i \le m$ and $1 \le j \le p$ is

$$(ab)_{ij} := a_{i1}b_{1j} + a_{i2}b_{2j} + \ldots + a_{in}b_{nj}.$$

↓ common rules for matrix multiplication ↓

Theorem
Suppose that A, B, and C are matrices (with sizes such that the multiplications and additions below make sense) and let $c \in \mathbb{R}$ be a scalar. Then
(a) $(AB)C = A(BC)$ (associativity);
(b) $A(B + C) = AB + AC$ (left distributivity);
(c) $(A + B)C = AC + BC$ (right distributivity);
(d) $c(AB) = (cA)B$.


B
$b_{1,1}$ $b_{1,2}$ $b_{1,3}$ $b_{1,4}$
$b_{2,1}$ $b_{2,2}$ $b_{2,3}$ $b_{2,4}$
A
$a_{1,1}$ $a_{1,2}$
$a_{2,1}$ $a_{2,2}$
$a_{3,1}$ $a_{3,2}$
AB
$= a_{1,1}b_{1,2} + a_{1,2}b_{2,2}$
$= a_{3,1}b_{1,4} + a_{3,2}b_{2,4}$

outdated

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Lecture 3 – Matrices and Matrix Operations (cont.), Inverses

slides 03 - matrices and matrix operations (cont.), inverses
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transposing matrices

Theorem (Theorem 1.3.4 of Johnston (2021))
Let A and B be matrices with sizes such that the operations below make sense and let $c \in \mathbb{R}$ be a scalar. Then
(a) $(A^T)^T = A$;
(b) $(A + B)^T = A^T + B^T$;
(c) $(AB)^T = B^TA^T$;
(d) $(cA)^T = cA^T$.

Definition (Definition 3.1.4 of Johnston (2021))
The *trace* of $A \in \mathbb{R}^{n\times n}$, denoted by $\operatorname{tr}(A)$, is the sum of its diagonal entries:

$$\operatorname{tr}(A) := a_{11} + a_{22} + \ldots + a_{nn}.$$

Theorem (Theorem 3.1.6 of Johnston (2021))
Suppose A and B are matrices whose sizes are such that the following operations make sense, and let c be a scalar.
(a) $\operatorname{tr}(A + B) = \operatorname{tr}(A) + \operatorname{tr}(B)$;
(b) $\operatorname{tr}(cA) = c\operatorname{tr}(A)$;
(c) $\operatorname{tr}(AB) = \operatorname{tr}(BA)$.

Definition (Definition 2.2.2 of Johnston (2021))
$A \in \mathbb{R}^{n\times n}$ is called *invertible* if there exists a matrix, which we denote by A^{-1} and call the *inverse* of A, such that

$$AA^{-1} = A^{-1}A = I_n.$$

use inverse matrix to solve

$A = \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix} \quad A^{-1} = \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}$

$A^{-1}A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ ← unige for A!

Theorem (Theorem 2.2.2 of Johnston (2021))
Let $A \in \mathbb{R}^{n\times n}$ be an invertible matrix, let $c \neq 0$, and let k be a positive integer. Then
(a) A^{-1} *is invertible and* $(A^{-1})^{-1} = A$;
(b) cA *is invertible and* $(cA)^{-1} = \frac{1}{c}A^{-1}$;
(c) A^T *is invertible and* $(A^T)^{-1} = (A^{-1})^T$;
(d) A^k *is invertible and* $(A^k)^{-1} = (A^{-1})^k$.

↕ common roles for invertable matrices

Theorem
A 2 × 2-matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

is invertible if and only if $ad - bc \neq 0$ and in this case

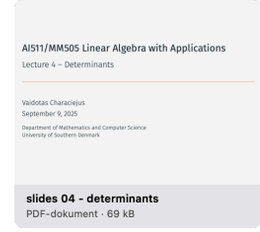
$$A^{-1} = \frac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Theorem (Theorem 2.2.3 of Johnston (2021))
Suppose that $A, B \in \mathbb{R}^{n\times n}$ are invertible. Then AB is also invertible, and

$$(AB)^{-1} = B^{-1}A^{-1}.$$

Socks & shoes

inversion of a matrix is only defined for a $\mathbb{R}^{n\times n}$ (square) matrices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Lecture 4 – Determinants

slides 04 - determinants
PDF-dokument · 69 kB

if A is invertable then A's rREF $= I_n$

find the inverse:

$A \xrightarrow{ERO} B$
use elementary row operation on both A & I_n
$I_n \xrightarrow{ERO} E$

note: left multiplication ↓ $EA = B$

example:

$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$

find inverse of A by bringing the augmented matrix

Gauss-Jordan

$$[A \mid I_3] = \left[\begin{array}{ccc|ccc} 1&2&3&1&0&0 \\ 2&5&3&0&1&0 \\ 1&0&8&0&0&1 \end{array}\right]$$

$$\xrightarrow{R_2 - 2R_1} \left[\begin{array}{ccc|ccc} 1&2&3&1&0&0 \\ 0&1&-3&-2&1&0 \\ 1&0&8&0&0&1 \end{array}\right]$$

$$\xrightarrow{R_3 - R_1} \left[\begin{array}{ccc|ccc} 1&2&3&1&0&0 \\ 0&1&-3&-2&1&0 \\ 0&-2&5&-1&0&1 \end{array}\right]$$

$$\xrightarrow{R_3 + 2R_2} \left[\begin{array}{ccc|ccc} 1&2&3&1&0&0 \\ 0&1&-3&-2&1&0 \\ 0&0&-1&-5&2&1 \end{array}\right]$$

$$\xrightarrow{1 - R_3} \left[\begin{array}{ccc|ccc} 1&2&3&1&0&0 \\ 0&1&-3&-2&1&0 \\ 0&0&1&5&-2&-1 \end{array}\right]$$

$$\xrightarrow{R_2 + 3R_3} \left[\begin{array}{ccc|ccc} 1&2&3&1&0&0 \\ 0&1&0&13&-5&-3 \\ 0&0&1&5&-2&-1 \end{array}\right]$$

$$\xrightarrow{R_1 - 3R_3} \left[\begin{array}{ccc|ccc} 1&2&0&-14&6&3 \\ 0&1&0&13&-5&-3 \\ 0&0&1&5&-2&-1 \end{array}\right]$$

$$\xrightarrow{R_1 - 2R_2} \left[\begin{array}{ccc|ccc} 1&0&0&-40&16&9 \\ 0&1&0&13&-5&-3 \\ 0&0&1&5&-2&-1 \end{array}\right]$$

A is invertable ⟺ $Ax = 0$ has a unige solution

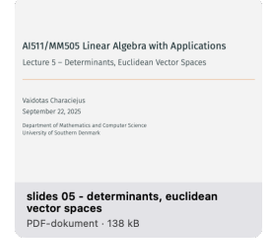
Step 1
$Ax = 0$
$BAx = B0$
$I_n x = 0$
$x = 0$

Step 2
$A^{-1}A = AA^{-1} = I_n$
$BA = I_n$
$\underbrace{BA}_{I_n}A^{-1} = I_nA^{-1}$
$B = A^{-1}$

for every $b \in \mathbb{R}^{n\times 1}$
where $A \in \mathbb{R}^{n\times n}$ & A is invertable
$Ax = b$
⇕
$x = A^{-1}b$

$Ax = b$
$A(A^{-1}b) = b$
$\underbrace{AA^{-1}}_{I_n}b = b$
$b = b$

$A\tilde{x} = b$
A^{-1}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Lecture 5 – Determinants, Euclidean Vector Spaces

slides 05 - determinants, euclidean vector spaces
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Some special matrices

$Ax = b \qquad A = PDP^{-1}$
$x = A^{-1}b \qquad A^2 = PD^2P^{-1}$
$AB = I_n \qquad A^{-1} = PD^{-1}P^{-1}$
$BA = I_n$

diagonal matrices

$\begin{bmatrix} 1&0&0 \\ 0&2&0 \\ 0&0&3 \end{bmatrix} = \operatorname{diag}(1,2,3)$

$\operatorname{diag}(a_1 \ldots a_n)\operatorname{diag}(b_1 \ldots b_n) = \operatorname{diag}(a_1b_1 \ldots a_nb_n)$

$D^{-1} = (d_1^{-1} \ldots d_n^{-1})$
(only if $d_i \neq 0$)
$D^2 = (d_1^2 \ldots d_n^2)$

triangular matrices

upper $= \begin{bmatrix} 1&2&3 \\ 0&4&5 \\ 0&0&6 \end{bmatrix}$

lower $= \begin{bmatrix} 1&0&0 \\ 2&3&0 \\ 4&5&6 \end{bmatrix}$

a) the transpose of a lower tri. matrix is upper
b) the product of 2 upper tri.-matrices are upper

Determinants

recall that for
$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$
$\det(A) = ad - bc$
if $\det(A) \neq 0$ then A is invertable

minors & cofactors

$A = \begin{bmatrix} a_{11} & \cancel{a_{12}} \\ \cancel{a_{21}} & a_{22} \end{bmatrix} \quad M_{11} = \det([a_{22}])$

$C_{ij} = (-1)^{i+j}M_{ij}$

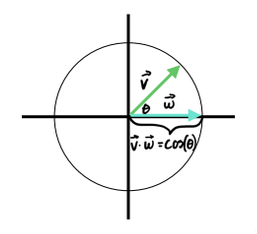
note that $\det(A + B) \neq \det(A) + \det(B)$ in all instances

if matrix A is triangular
$\det(A) = \prod_{i=1}^{n} a_{ii}$

for matrix $A \in \mathbb{R}^{n\times n}$
(a) by multiplying a row with $k \in \mathbb{R}$, then $\det(B) = k \cdot \det(A)$
(b) interchanging two rows, then $\det(B) = -\det(A)$
(c) adding a multiple one row to another, then $\det(B) = \det(A)$

A matrix A is invertible if and only if $\det(A) \neq 0$

Vectors



observe that
$\vec{v}\cdot\vec{w} = \vec{v}^T\vec{w} = \vec{v}\,\vec{w}^T$
$\|\vec{v}\| = \sqrt{v_1^2 + v_2^2 \ldots v_n^2}$
euclidean norm

for $\vec{v} \in \mathbb{R}^n$ & $c \in \mathbb{R}$
(a) $\|c\vec{v}\| = |c|\|\vec{v}\|$

for any $v = 0$ vectors
$\vec{v} = \|\vec{v}\|\vec{u}$
where $\vec{u}$ is a unit vector

the standard basis of in $\mathbb{R}^n$

$e_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix},\ e_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \ldots, e_n \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}.$

and so...
any vector in $\mathbb{R}^n$ can be written in terms of the vectors $e_1, e_2, \ldots, e_n$ in the following way

$v = (v_1e_1 + v_2e_2 + \ldots + v_ne_n)$

Propeties of matrix transformations

linear transformations
satisfies the following properties
(a) $T(v + w) = T(v) + T(w)$ for all vectors $v, w \in \mathbb{R}^n$

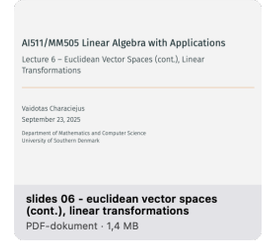
transformation is linear if
scaling then mapping =
mapping then scaling

Standard matrix of a linear transformation
a function $T: \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation if and only if there exists a matrix $[T] \in \mathbb{R}^{m\times n}$ such that
$T(v) = [T]v$ for all $v \in \mathbb{R}^n$

note:
$[T] = [T(e_1)\ T(e_2) \ldots T(e_n)]$

example
$T(v_1, v_2) = (v_1 + 2v_2,\ 3v_1 + 4v_2)$
$T(e_1) = T(1,0) = (1,3)$
$T(e_2) = T(0,1) = (2,4)$
$[T] = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$
$[T]\cdot\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} v_1 & 2v_2 \\ 3v_1 & 4v_2 \end{bmatrix}$

Definition 1.4.1 of Johnston (2021)
A *linear transformation* is a function $T : \mathbb{R}^n \to \mathbb{R}^m$ that satisfies the following two properties:
(a) $T(v + w) = T(v) + T(w)$ for all vectors $v, w \in \mathbb{R}^n$;
(b) $T(cv) = cT(v)$ for all vectors $v \in \mathbb{R}^n$ and all scalars $c \in \mathbb{R}$.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Lecture 6 – Euclidean Vector Spaces (cont.), Linear Transformations

slides 06 - euclidean vector spaces (cont.), linear transformations
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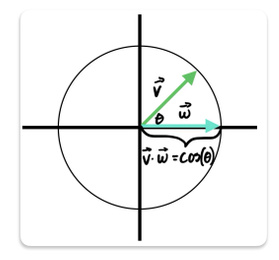
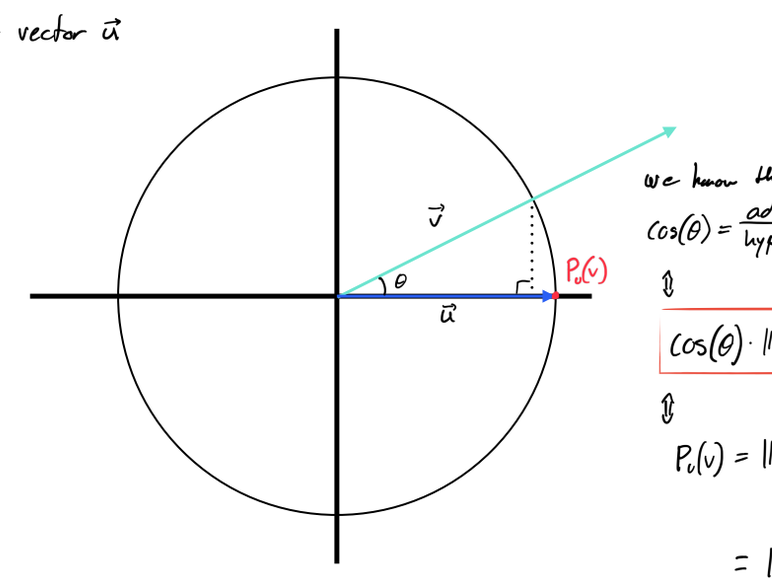
Orthogonal projection onto the direction of a unit vector $\vec{u}$



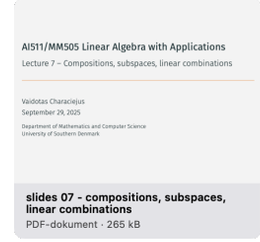


we know that
$\cos(\theta) = \frac{adj}{hyp} = \frac{\|P_u(v)\|}{\|v\|}$
⇕
$\boxed{\cos(\theta)\cdot\|v\| = \|P_u(v)\|}$
⇕
$P_u(v) = \|v\|\cdot\cos(\theta)\cdot u$
$= \|v\|\cdot\frac{v\cdot u}{\|u\|\,\|v\|}\cdot u$
$= (v\cdot u)\cdot u$
$= u^Tvu$
$= uu^Tv$

Dot product

$\frac{w\cdot v}{\|w\|\cdot\|v\|} = \cos(\theta)$

so for unit vectors
$w\cdot v = \cos(\theta)$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Lecture 7 – Compositions, subspaces, linear combinations

slides 07 - compositions, subspaces, linear combinations
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irrelevant

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Lecture 8 – Subspaces, Linear Combinations, Bases and Coordinates, Dimension

slides 08 - subspaces, linear combinations, bases and coordinates, dimension
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Subspaces

Definition 2.3.1 of Johnston (2021)
A *subspace* of $\mathbb{R}^n$ is a non-empty set $\mathcal{S}$ of vectors in $\mathbb{R}^n$ with the properties that
(a) if $v, w \in \mathcal{S}$, then $v + w \in \mathcal{S}$;
(b) if $v \in \mathcal{S}$ and $c \in \mathbb{R}$, then $cv \in \mathcal{S}$.

- (a) ensures that subspaces are "flat".
- (b) makes it so that they are "infinitely long".
- Observe that every subspace contains the zero vector (think why this is the case).

Definition 2.3.2 of Johnston (2021)
Suppose $A \in \mathbb{R}^{m\times n}$.
(a) The *range* of A is the subspace of $\mathbb{R}^m$, denoted by $\operatorname{range}(A)$, that consists of all vectors of the form Ax.
(b) The *null space* of A is the subspace of $\mathbb{R}^n$, denoted by $\operatorname{null}(A)$, that consists of all solutions x of the system of linear equations $Ax = 0$.

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Lecture 9 – Coordinates, Dimension

slides 09 - coordinates, dimension
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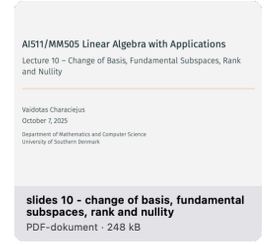
irrelevant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Lecture 10 – Change of Basis, Fundamental Subspaces, Rank and Nullity

slides 10 - change of basis, fundamental subspaces, rank and nullity
PDF-dokument · 248 kB

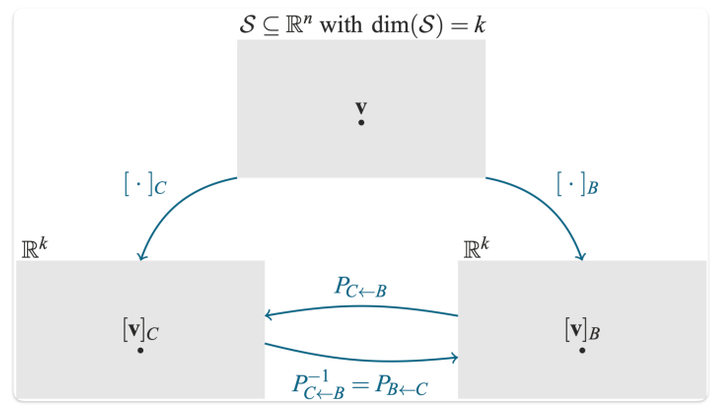
Chang-of-basis matrix

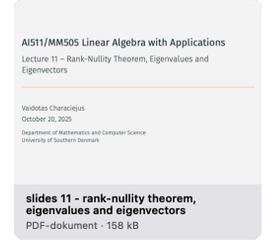
Definition 3.1.2 of Johnston (2021)
Let $\mathcal{S}$ be a subspace of $\mathbb{R}^n$ with bases $B = \{v_1, v_2, \ldots, v_k\}$ and C. The *change-of-basis matrix* from B to C, denoted by $P_{C\leftarrow B}$, is the $k \times k$ matrix whose columns are the coordinate vectors $[v_1]_C, [v_2]_C, \ldots, [v_k]_C$:

$$P_{C\leftarrow B} := \big[[v_1]_C \ \ [v_2]_C \ \ \ldots \ \ [v_k]_C\big].$$

Theorem 3.1.3 of Johnston (2021)
Suppose B and C are bases of a subspace $\mathcal{S}$ of $\mathbb{R}^n$, and let $P_{C\leftarrow B}$ be the change-of-basis matrix from B to C. Then
(a) $P_{C\leftarrow B}[v]_B = [v]_C$ for all $v \in \mathcal{S}$, and
(b) $P_{C\leftarrow B}$ is invertible and $P_{C\leftarrow B}^{-1} = P_{B\leftarrow C}$.
Furthermore, $P_{C\leftarrow B}$ is the unique matrix with property (a).

for $P_{C\leftarrow B} \in \mathbb{R}^{k\times k}$
$P_{C\leftarrow B}[v]_B = [v]_C$
also..
$P_{C\leftarrow B}^{-1} = P_{B\leftarrow C}$


$\mathcal{S} \subseteq \mathbb{R}^n$ with $\dim(\mathcal{S}) = k$
v
$[\cdot]_C$
$[\cdot]_B$
$\mathbb{R}^k$
$\mathbb{R}^k$
$[v]_C$
$[v]_B$
$P_{C\leftarrow B}$
$P_{C\leftarrow B}^{-1} = P_{B\leftarrow C}$


AI511/MM505 Linear Algebra with Applications
Lecture 11 – Rank-Nullity Theorem, Eigenvalues and Eigenvectors

slides 11 - rank-nullity theorem, eigenvalues and eigenvectors
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Lecture 12 – Eigenvalues and Eigenvectors (cont.), Diagonalisation

slides 12 - eigenvalues and eigenvectors (cont.), diagonalisability
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